Simulation Arena

Deal Approval & Margin Balancing

Approve 0% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $361.92M of expected profit.

Seed 42 | Run run_b1c17e4e2600 | 170,000 scenarios tested | Monte Carlo simulation with an empirica… | 22 Sep 2026, 07:26

Same seed, same numbers - this run can be reproduced exactly.

How this works

For each discount level we ran thousands of 'what-if' worlds. In each one the buyer's odds of saying yes wobble a little, and your delivery cost wobbles a little. We then ask the only question that matters: if you offered this discount again and again, how much profit would you bank on average? That is the expected profit - the chance of winning multiplied by the profit you keep. Because those worlds are a sample, we also work out how much the answer itself would wobble if we ran it again. Any discount whose expected profit is within twice that wobble of the best - or within 1% of the best profit, whichever is wider - is treated as giving the same answer. That is the range where you have genuine room to negotiate.

The numbers that matter

BEST DISCOUNT TO OFFER
0%
This is where the extra wins are still worth more than the margin given away.
vs rep asked for 25%

EXPECTED PROFIT AT THAT DISCOUNT
$43.67M
Average profit if you made this call many times over.

DISCOUNT RANGE YOU CAN APPROVE WITHOUT LOSING MONEY
0%
Only one discount stands out - the model cannot tell any neighbouring discount apart from it. The model's own margin of error here is about $3.50M, so treat this as the best single number - the next discount along already costs real money.
vs rep asked for 25% - outside the range

WHERE IT STARTS TO GENUINELY COST YOU
2%
At 2% the expected profit falls $18.14M below the best. That gap is wider than the model's own margin of error, so the loss is real.
vs costs about $18.14M

EXPECTED PROFIT IF YOU CAVE IN
-$318.25M
What 25% actually earns you.
vs -$361.92M vs the optimum

PROFIT GIVEN AWAY
$361.92M
Buying +48 points of win probability costs you this much.

ODDS OF WINNING AT THE BEST DISCOUNT
37.0%
The chance the deal lands.

WORST CASE (1-IN-20)
-$5.76M
In an unlucky run of 20 deals, this is roughly the profit you would see.

How we got there

1. The two forces fighting each other

Every point of discount does two things: it makes the buyer a bit more likely to say yes, and it takes $19.67M straight off your profit. Early on the first effect wins. Later, the second one does.

Evidence: Cost to deliver $1,849.36M on $1,966.78M of list value

2. What the buyer's behaviour looks like

They start around 35.0% likely to buy at full price and never become certain - the curve flattens near 95.0%. Around 17% discount is where they are most persuadable.

Evidence: Curve source: 2000 of your past deals

3. The sweet spot

0% discount gives the highest expected profit at $43.67M, with a 37.0% chance of winning.

Evidence: peak of 17 levels tested

4. The range where the answer is the same

0% is the only discount that stands out; the model cannot tell any neighbouring discount apart from it. The next level along already costs $18.14M.

Evidence: margin of error $3.50M; safe range 0%

5. What the requested discount really costs

Moving from 0% to 25% buys +48 points of win probability but costs $361.92M of expected profit - more than the model's margin of error, so the loss is real.

Evidence: -$318.25M vs $43.67M

Charts

Where profit peaks

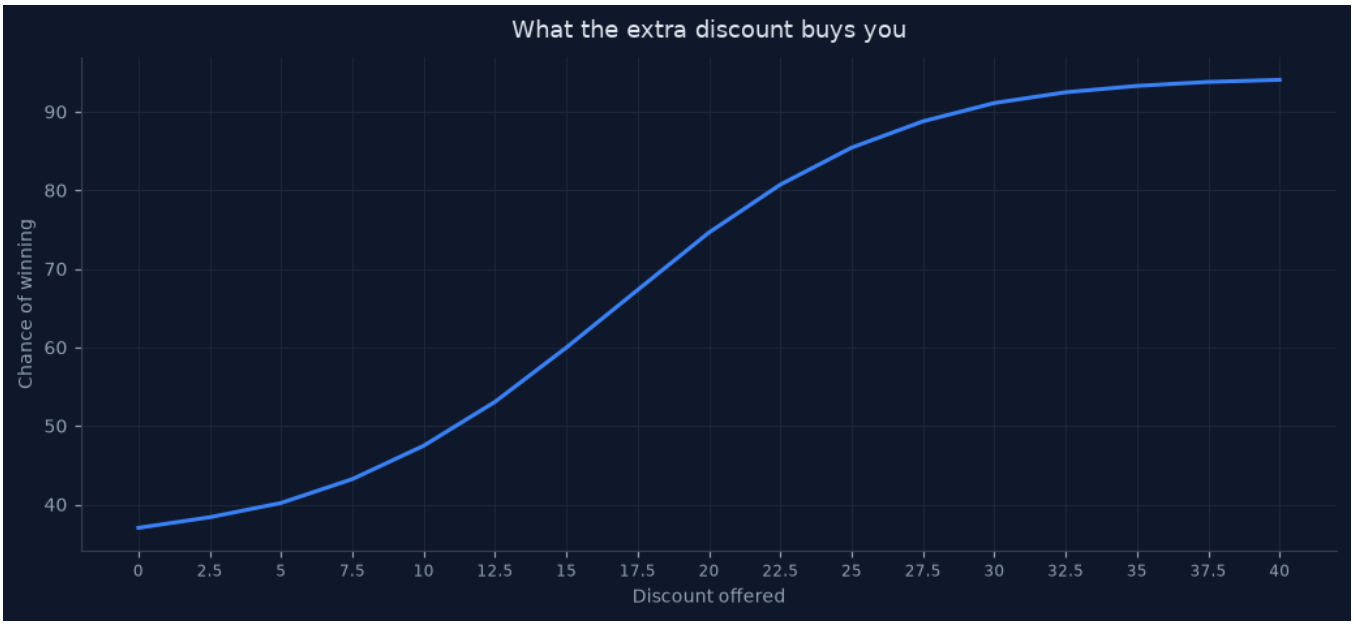
Everything from 0% gives the same answer within the model's margin of error; the peak is 0%



What this shows: Grey is what you keep per won deal - it only falls. Green is what you actually expect to earn - it rises, peaks, then falls. The amber dashed line is the best profit minus the model's own margin of error: every discount whose green line stays above it is the same answer, which is the 0% range. Past that you are genuinely paying for the win.

What the extra discount buys you

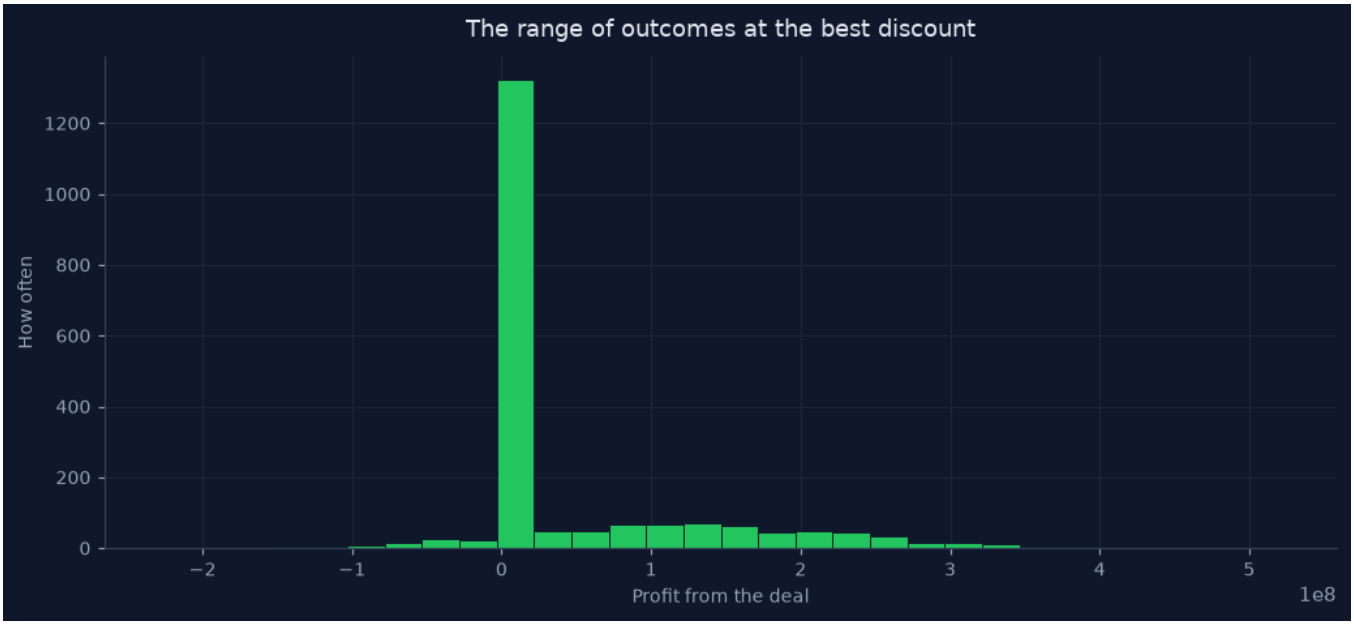
Curve learned from 2000 of your past deals



What this shows: Past the knee of this curve, extra discount buys almost no extra certainty - but keeps costing margin. That is the trap.

The range of outcomes at the best discount

10,000 simulated worlds at 0% discount



What this shows: The cluster at zero is the deals you lose. The spread on the right is the profit you make when you win.

Tables

Every discount level tested

Expected profit = chance of winning x profit kept. Highlighted row is the winner.

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
0.0%	$1,966,776,723	$117,211,950	37.0%	$43,674,006	-$5,756,224	86% chance
2.5%	$1,917,607,305	$69,278,431	38.4%	$25,531,019	-$56,069,153	93% chance
5.0%	$1,868,437,887	$18,988,644	40.2%	$7,345,252	-$111,484,855	97% chance
7.5%	$1,819,268,469	-$30,780,999	43.2%	-$13,056,522	-$163,972,440	99% chance
10.0%	$1,770,099,051	-$79,731,296	47.5%	-$36,938,001	-$210,631,635	100% chance
12.5%	$1,720,929,633	-$125,150,024	53.1%	-$65,892,197	-$267,075,099	100% chance

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
15.0%	$1,671,760,215	-$176,997,850	60.0%	-$106,805,713	-$331,686,051	100% chance
17.5%	$1,622,590,796	-$224,023,881	67.3%	-$150,441,713	-$384,368,050	100% chance
20.0%	$1,573,421,378	-$274,622,469	74.6%	-$205,479,947	-$442,266,101	100% chance
22.5%	$1,524,251,960	-$324,674,733	80.7%	-$262,628,459	-$496,172,899	100% chance
25.0%	$1,475,082,542	-$374,472,262	85.4%	-$318,245,391	-$548,124,994	100% chance
27.5%	$1,425,913,124	-$424,374,069	88.8%	-$375,387,236	-$598,381,482	100% chance
30.0%	$1,376,743,706	-$475,225,167	91.1%	-$432,985,188	-$656,003,419	100% chance
32.5%	$1,327,574,288	-$522,362,017	92.5%	-$483,021,041	-$703,821,438	100% chance
35.0%	$1,278,404,870	-$572,751,351	93.3%	-$535,377,436	-$750,335,917	100% chance
37.5%	$1,229,235,452	-$618,378,624	93.8%	-$579,725,765	-$794,912,915	100% chance
40.0%	$1,180,066,034	-$669,362,260	94.1%	-$630,373,810	-$847,754,142	100% chance

Row 1 maximises expected profit.

The decision in one line

What changes if you approve the discount being asked for.

Column 1	Best (0%)	Asked (25%)	Difference
Discount	0.0%	25.0%	+25.0 pts
Win chance	37.0%	85.4%	+48.4 pts
Profit if won	$117,211,950	-$374,472,262	-$491,684,212
Expected profit	$43,674,006	-$318,245,391	-$361,919,397
Range where the answer is the same	0%	outside it	margin of error $3.50M

The range where the answer is the same

Every discount we tested, how far it falls short of the best, and whether that shortfall is real money or just the model's own margin of error.

Discount	Expected profit	Short of the best	Inside the range?	What it means
0.0%	$43,674,006	$0	Inside	safe to approve
2.5%	$25,531,019	$18,142,988	Outside	costs you $18.14M
5.0%	$7,345,252	$36,328,754	Outside	costs you $36.33M
7.5%	-$13,056,522	$56,730,528	Outside	costs you $56.73M
10.0%	-$36,938,001	$80,612,007	Outside	costs you $80.61M
12.5%	-$65,892,197	$109,566,203	Outside	costs you $109.57M
15.0%	-$106,805,713	$150,479,719	Outside	costs you $150.48M
17.5%	-$150,441,713	$194,115,720	Outside	costs you $194.12M
20.0%	-$205,479,947	$249,153,954	Outside	costs you $249.15M
22.5%	-$262,628,459	$306,302,465	Outside	costs you $306.30M
25.0%	-$318,245,391	$361,919,397	Outside	costs you $361.92M
27.5%	-$375,387,236	$419,061,242	Outside	costs you $419.06M
30.0%	-$432,985,188	$476,659,194	Outside	costs you $476.66M
32.5%	-$483,021,041	$526,695,047	Outside	costs you $526.70M
35.0%	-$535,377,436	$579,051,442	Outside	costs you $579.05M
37.5%	-$579,725,765	$623,399,771	Outside	costs you $623.40M
40.0%	-$630,373,810	$674,047,816	Outside	costs you $674.05M

Highlighted rows are the range where the answer is the same (0%) - within about $3.50M of the best.

Recommended discount per deal (top 25)

The same logic applied to each deal individually.

Deal	Customer	Value	Margin	Best discount	Expected profit
Q-000000846	-	$11,159,991	-0.8%	0%	-$32,243
Q-000001769	-	$10,909,168	7.3%	0%	$294,575
Q-000001338	-	$10,907,795	30.3%	0%	$1,226,233
Q-000001616	-	$9,335,947	-17.8%	0%	-$616,922
Q-000001456	-	$8,959,167	8.7%	0%	$288,712
Q-000000631	-	$8,552,832	12.4%	0%	$391,567
Q-000000506	-	$8,441,267	22.5%	0%	$703,818
Q-000000139	-	$8,206,052	11.4%	0%	$346,814
Q-000001077	-	$7,804,111	-0.6%	0%	-$17,922
Q-000000248	-	$7,137,207	8.8%	0%	$233,699
Q-000001345	-	$7,122,317	4.8%	0%	$126,895
Q-000001195	-	$6,967,127	10.8%	0%	$278,453
Q-000000479	-	$6,946,073	4.8%	0%	$123,240
Q-000000030	-	$6,524,061	20.5%	0%	$496,359
Q-000001446	-	$6,436,324	10.9%	0%	$259,861
Q-000001973	-	$6,324,179	27.0%	0%	$632,946
Q-000001251	-	$6,053,921	15.8%	0%	$353,179
Q-000001411	-	$6,000,113	22.2%	0%	$494,056
Q-000000553	-	$5,946,861	13.0%	0%	$286,578
Q-000000226	-	$5,942,804	33.4%	0%	$734,336
Q-000000244	-	$5,703,007	16.4%	0%	$345,804
Q-000001053	-	$5,701,775	-23.8%	0%	-$502,226
Q-000000676	-	$5,636,582	-0.5%	0%	-$10,648
Q-000000960	-	$5,506,333	-12.8%	0%	-$260,862
Q-000001660	-	$5,472,739	14.2%	0%	$288,664

What we recommend

Approve 0% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $361.92M of expected profit.

Risk High 100 out of 100 | Confidence: High

Why

• Only 0% stands out from the rest - the model cannot tell any neighbouring discount apart from it.

• Expected profit peaks at 0% ($43.67M).

• The requested 25% delivers -$318.25M - $361.92M less than the peak, which is more than the model's margin of error.

• Win probability only improves +48 points for that giveaway.

What to do

1. Approve 0% as the standard concession - it is the only discount the model can tell apart from the rest.

2. Above 0% (the top of the safe range), require VP sign-off: the 25% being asked for costs about $14.48M per point of discount.

3. At 25% there is a 100% chance of breaching the 12.0% margin floor - add a volume or term commitment to earn the deeper discount.

The trade-off: You are trading $361.92M of expected profit for +48 points of win probability.

What we assumed and what to watch

Worth knowing before you act

- At 25% discount there is a 100% chance the deal actually costs you money to deliver even if you win it.

Assumptions

- List value $1,966,776,723; cost to deliver $1,785,978,353 at the 9.2% standard margin your file implies, plus $63,383,412 of consumption effects on top, giving a total cost base of $1,849,361,764 and an effective 6.0% margin once those effects are included.
- Consumption effects: 0.2% of value is consumed as unmetered burst you still pay for; 4 months of rollover defers revenue.
- Win probability curve: from 35.0% at full price toward a ceiling of 95.0%, halfway at 16.7% discount (source: 2000 of your past deals).
- Win probability uncertainty: plus or minus 5 points.
- Cost volatility: plus or minus 6% of cost.
- Margin floor: 12.0%.
- Two discounts are treated as giving the same answer when the gap between their expected profits is smaller than twice the model's own margin of error at both discounts added together (about $3.50M at the best one), or smaller than 1% of the best profit - whichever is larger.
- The range where the answer is the same is 0%, taken from the 17 discount levels we tested.
- 10,000 simulations per discount level; seed 42 (re-running gives identical numbers).
- Deal / quote ID -> your 'deal_id' column
- Customer name -> not in your file; not used
- Deal value (list price) -> your 'deal_value' column
- Cost to deliver -> your 'cost' column
- Standard margin (%) -> your 'margin_pct' column
- Discount given (%) -> your 'discount_pct' column
- Won? (1/0) -> your 'won' column
- Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column
- Unmetered burst used -> your 'burst_allowance_pct' column
- Rollover window -> your 'rollover_window_months' column

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	Won? (1/0) -> your 'won' column
0.00s	prepare	Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column
0.00s	prepare	Unmetered burst used -> your 'burst_allowance_pct' column
0.00s	prepare	Rollover window -> your 'rollover_window_months' column
0.08s	prepare	Read 2,000 records into the model.
0.08s	simulate	Running the simulation…
0.09s	read	Reading the deal you gave us...
0.10s	read	Found 2000 deal(s) worth $1,966.78M at an average 9.2% margin.

Time	Stage	What happened
0.10s	prepare	Cost to deliver is about $1,785.98M.
0.11s	prepare	Added the consumption effects from your file: $63.38M of extra cost to serve, taking the cost base to $1,849.36M.
0.11s	prepare	Working out how much a discount really moves the buyer...
0.16s	prepare	Learned the buyer's behaviour from 2000 past deals (they flip hardest around 16.7% discount).
0.16s	simulate	Testing 17 discount levels, 10,000 outcomes each (170,000 scenarios in total)...
0.17s	simulate	Simulated 11% of the discount levels...
0.17s	simulate	Simulated 23% of the discount levels...
0.17s	simulate	Simulated 35% of the discount levels...
0.17s	simulate	Simulated 47% of the discount levels...
0.18s	simulate	Simulated 58% of the discount levels...
0.18s	simulate	Simulated 70% of the discount levels...
0.18s	simulate	Simulated 82% of the discount levels...
0.18s	simulate	Simulated 94% of the discount levels...
0.18s	simulate	Simulated 100% of the discount levels...
0.18s	summarize	All discount levels tested. Looking for the sweet spot...
0.18s	summarize	Checked how much the answer would wobble if we ran it again - the range where the answer is the same is 0%.
0.19s	summarize	Done. Best discount is 0% ($43.67M expected profit); the range where the answer is the same is 0%.